

Placement Empowerment Program
Cloud Computing and DevOps Centre

Automate File Copying with a Script

Create a script to copy files from one folder to another automatically.

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Introduction and Overview

This Proof of Concept (POC) aims to provide a thorough understanding of automating the file copying process using a script. The process involves creating a script that automatically copies files from one folder to another without manual intervention. By utilizing a batch script, you will gain hands-on experience in automating routine tasks and improving workflow efficiency. This POC will also introduce you to the concept of task automation on Windows systems, which is an essential skill for IT professionals, developers, and system administrators. By the end of this exercise, you will be able to automate file management tasks, reducing the need for repetitive manual processes and ensuring a more streamlined and efficient system.

Objective

The primary objective of this project is to:

1. **Create an Automated File Copying Script** – Learn how to write a script that can copy files from one folder to another automatically using simple commands.
2. **Understand File Management Automation** – Gain knowledge of task automation and explore how it can simplify the management of files and directories.

Importance of Automating File Copying

Automating file copying plays a crucial role in optimizing workflow and ensuring that important data is consistently backed up or moved without human intervention. It is especially useful for tasks that involve managing large sets of files, such as backups, updates, and data migration. Some of the key advantages of automating file copying include:

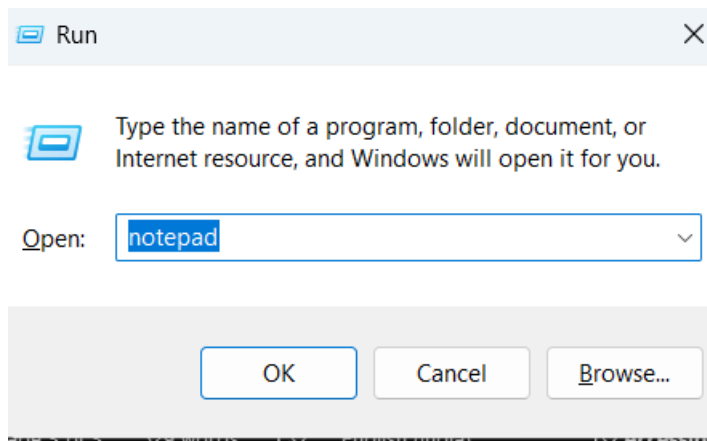
- **Time Efficiency** – By automating repetitive tasks like file copying, developers and system administrators can save time and focus on more complex tasks, improving overall productivity.
- **Consistency and Accuracy** – Automated scripts ensure that the file copying process is done the same way every time, reducing the risk of errors or missed steps.
- **Ease of Implementation** – Writing a batch script for file copying is a simple yet powerful solution that can be easily adapted to different scenarios and used on any Windows-based system.

By completing this POC, you will gain valuable experience in automating basic file management tasks, improving your ability to optimize and streamline workflows, and understanding the power of scripting for system administration.

Steps for execution:

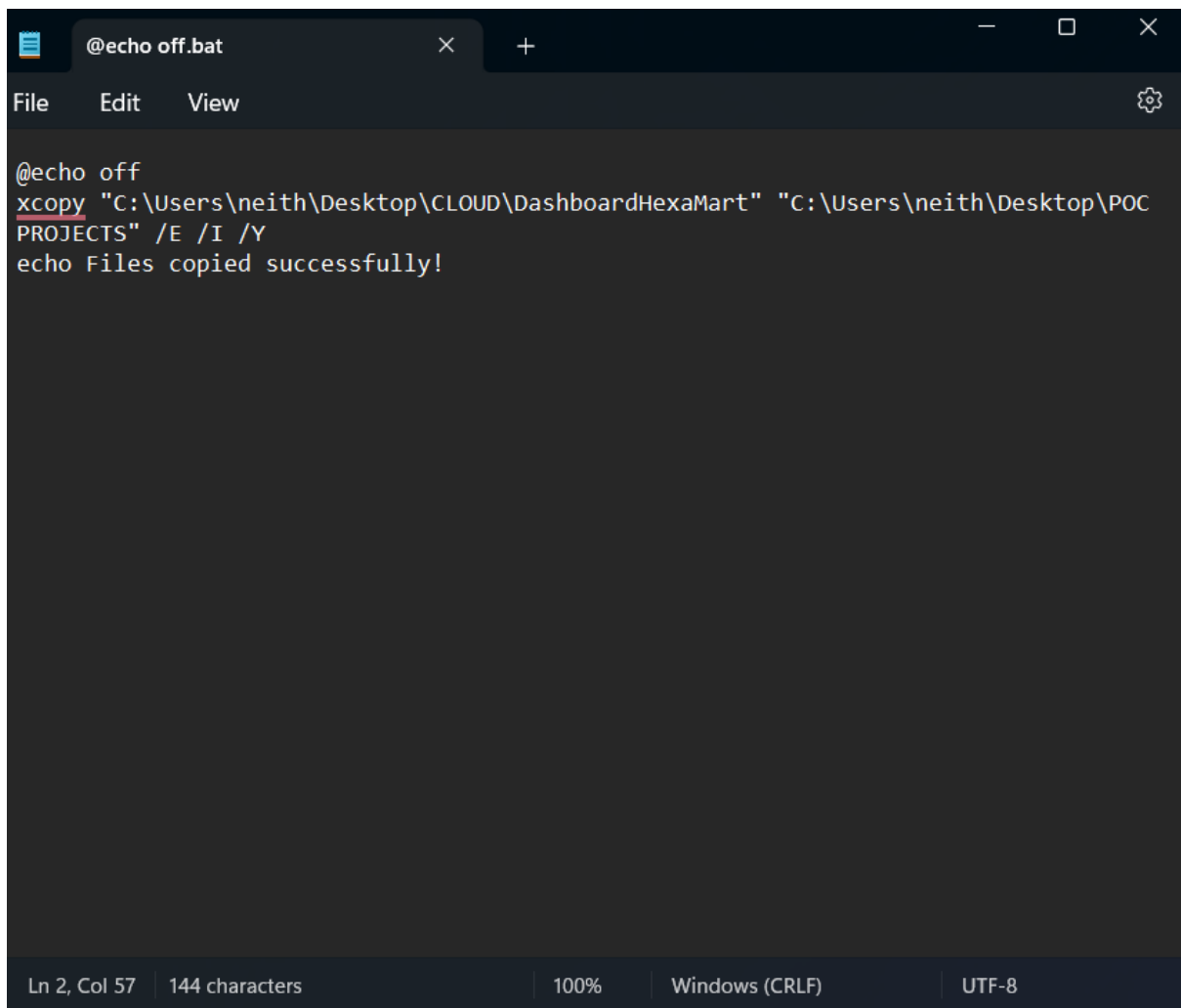
Step 1: Open Notepad

I opened Notepad on my Windows computer.



Step 2: Write the Script

I wrote a batch script using the xcopy command to copy files from one folder to another. I specified the source folder where the files are located and the destination folder where the files should be copied. I also included necessary flags to ensure all files and subdirectories were copied, and existing files were overwritten without prompts.



The screenshot shows a text editor window with the title bar "@echo off.bat". The menu bar includes "File", "Edit", and "View". The text content is as follows:

```
@echo off
xcopy "C:\Users\neth\Desktop\CLOUD\DashboardHexaMart" "C:\Users\neth\Desktop\POC
PROJECTS" /E /I /Y
echo Files copied successfully!
```

The status bar at the bottom indicates "Ln 2, Col 57", "144 characters", "100%", "Windows (CRLF)", and "UTF-8".

Step 3: Save the File as a Batch Script

I clicked **File** → **Save As**, changed "**Save as type**" to "**All Files**", and named the file with a .bat extension (e.g., echo_off.bat).



@echo off.bat	17-02-2025 22:40	Windows Batch File	1 KB
backup_git.bat	16-02-2025 21:27	Windows Batch File	1 KB
index.html	16-02-2025 14:22	Microsoft Edge HT...	1 KB
Mywebsite.html	16-02-2025 14:20	Microsoft Edge HT...	2 KB
Mywebsite.txt	16-02-2025 21:15	Text Document	2 KB
README.md	16-02-2025 14:14	Markdown Source ...	1 KB

Step 4: Run the Batch File

I located the saved .bat file and double-clicked it to execute the script. The command prompt opened, and the script ran, copying the files as intended. If needed, I ran it as an Administrator to avoid permission issues.

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Slide1.PNG	11-02-2025 23:29	PNG File	136 KB

Key Takeaways

The **course outcome** for automating file copying with a batch script would typically be:

1. **Skill in Automation:** The ability to automate repetitive tasks, such as copying files, using simple scripts without manual intervention.
2. **Understanding Batch Scripting:** Gaining a practical understanding of batch scripting to perform file operations and basic system tasks on Windows systems.
3. **Task Scheduling:** Developing the ability to automate the execution of batch scripts at specified times or events using Windows Task Scheduler.
4. **Command Line Proficiency:** Becoming comfortable working with the command line interface (CLI) for executing scripts and performing file management tasks.
5. **Error Handling and Efficiency:** Learning to include features such as file overwriting, handling subdirectories, and ensuring the script runs smoothly without errors.